Hermes Insertion-Site Identification by Nested Arbitrary PCR

Objective
To identify the genomic insertion site of the Hermes transposon in individual *Candida glabrata* mutants using two rounds of nested arbitrary PCR followed by Sanger sequencing.

The method is adapted from the arbitrary PCR strategy described by Das et al. (2005) and Martínez-García et al. (2014). The published cycling architecture is retained, while denaturation and extension conditions are adapted for **OneTaq Master Mix**.

Scale as needed

PCR1 total volume: 50 µL

Template: 20 ng genomic DNA

alternatively, 5 µL cell lysate

PCR2 total volume: 50 µL

PCR2 template: 1 µL PCR1 product (or 40 ng purified PCR1 DNA)

Primer Architecture

PCR1

Hermes-specific outer primer (HR-1)
AATACTTGCACTCAAAAGGCTTGACACC

Arbitrary primer (ARB6)
GGCACGCGTCGACTAGTACNNNNNNNNNNACGCC

PCR1 uses the Hermes-specific outer primer together with the degenerate ARB6 primer to amplify unknown genomic DNA adjacent to the Hermes insertion.

PCR2

Hermes-specific nested primer (HR-2)
ACACTTGTGCTTATCTATGTGGC

Nested arbitrary primer (ARB2)
GGCACGCGTCGACTAGTAC

PCR2 uses **1 µL of the PCR1 product** as template. HR-2 is nested relative to HR-1, while ARB2 recognizes the defined 5′ sequence introduced by ARB6 during PCR1.

PCR1 - Arbitrary Priming

Template: *C. glabrata* genomic DNA from an individual Hermes mutant
Primers: HR-1 + ARB6
Polymerase: OneTaq Master Mix

Cycling Conditions

Step	**Temperature**	**Time**	**Cycles**
Initial denaturation	94°C	30 sec	1
Denaturation	94°C	30 sec	
Low-stringency annealing	30°C	30 sec	6
Extension	68°C	90 sec	
Denaturation	94°C	30 sec	
Annealing	45°C	30 sec	30
Extension	68°C	90 sec	
Final extension	68°C	5 min	1
Hold	4°C	∞	

Thermocycler Program

94°C 30 sec

94°C 30 sec

30°C 30 sec × 6 cycles

68°C 90 sec

94°C 30 sec

45°C 30 sec × 30 cycles

68°C 90 sec

68°C 5 min

4°C HOLD

Note: The 30°C annealing temperature is intentionally retained from the published arbitrary-PCR method. This low-stringency step allows the degenerate 3′ region of ARB6 to establish genomic priming sites and should not be increased to the standard OneTaq annealing range.

PCR2 - Nested Amplification

Template: 1 µL PCR1 product
Primers: HR-2 + ARB2
Polymerase: OneTaq Master Mix

Cycling Conditions

Step	Temperature	Time	Cycles
Initial denaturation	94°C	30 sec	1
Denaturation	94°C	30 sec	
Annealing	52°C	30 sec	30
Extension	68°C	90 sec	
Final extension	68°C	5 min	1
Hold	4°C	∞	

Thermocycler Program

94°C 30 sec

94°C 30 sec

52°C 30 sec × 30 cycles

68°C 90 sec

68°C 5 min

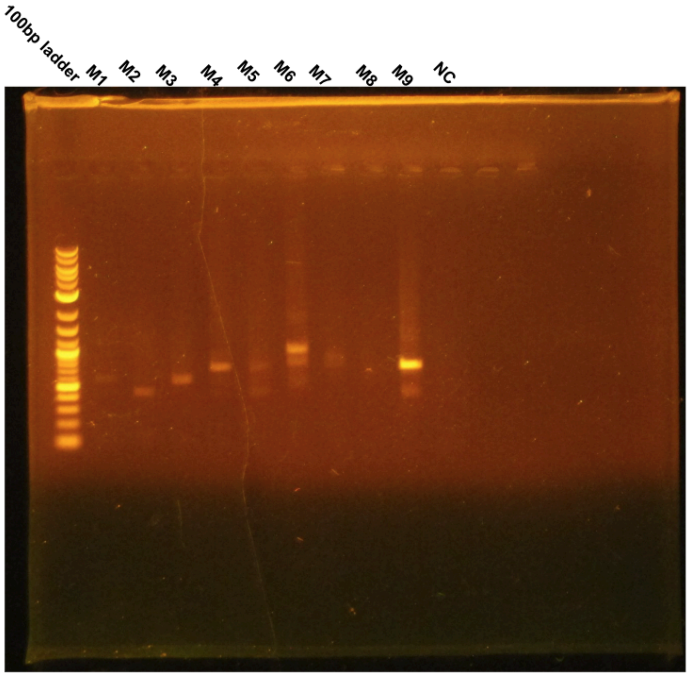
4°C HOLD

PCR2 Product Assessment
Run an aliquot of the PCR2 product on an agarose gel.

Arbitrary PCR products may appear as a **smear or multiple products rather than a single discrete band**. Therefore, the presence of a smear does not necessarily indicate PCR failure.

PCR2 products suitable for sequencing will be purified prior to Sanger sequencing.

Sanger Sequencing
Hermes-specific nested primer (HR-2)
ACACTTGTGCTTATCTATGTGGC



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PCR1

Hermes-specific outer primer (HR-1)
AATACTTGCACTCAAAAGGCTTGACACC

Arbitrary primer (ARB6)
GGCACGCGTCGACTAGTACNNNNNNNNNNACGCC

PCR2

Hermes-specific nested primer (HR-2)
ACACTTGTGCTTATCTATGTGGC

Nested arbitrary primer (ARB2)
GGCACGCGTCGACTAGTAC

Sanger Seq
Hermes-specific nested primer (HR-2)
ACACTTGTGCTTATCTATGTGGC

Conclusion
PCR adaptation appeared successful. Single bands mostly recovered (no concerning smearing). The products were purified and submitted to the seq core for sanger seq

References

Das S, Noe JC, Paik S, Kitten T. An improved arbitrary primed PCR method for rapid characterization of transposon insertion sites. *Journal of Microbiological Methods*. 2005;63(1):89–94.

Martínez-García E, Aparicio T, de Lorenzo V, Nikel PI. New transposon tools tailored for metabolic engineering of Gram-negative microbial cell factories. *Frontiers in Bioengineering and Biotechnology*. 2014;2:46.

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